SAI AMAN GOPISETTI

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PROFESSIONAL SUMMARY

Battery materials engineer with 4+ years of hands-on experience in carbon-based anode development for Li-ion batteries, electrolyte evaluation, coin cell assembly, and electrochemical characterization. Developed novel electrode chemistries from concept through cell-level validation, achieving significant energy density and voltage improvements over conventional Li-ion coin cells in half-cell validation. Named inventor on granted U.S. Patent (No. 12,308,433) with three additional filings in pipeline. Experienced in coordinating R&D activities across external research institutions and managing iterative development under resource constraints. Strong foundation in materials science, battery aging analysis, degradation mechanisms, and data analysis (MATLAB).

CORE COMPETENCIES

Battery Cell Development

- Carbon-based anode material development, surface modification, and electrode fabrication
- Electrolyte selection and evaluation (ionic liquid, carbonate-based, additives)
- Coin cell assembly and half-cell testing (2032 format)
- Battery aging and degradation analysis (capacity fade, impedance, SEI, SoH)
- Graphitic structure analysis and material optimization

Characterization & Testing

- SEM, XRD, FTIR, Raman, XPS, BET
- Electrochemical testing (galvanostatic cycling, CV, EIS, dQ/dV)
- Specification development and qualification
- Design of experiments (DOE)
- Failure analysis and root cause investigation

Fabrication & Processing

- Pulsed Laser Deposition (PLD)
- Cleanroom fabrication (NUANCE, NUfab)
- Inert atmosphere (N2) glovebox operation
- Thin film fabrication and testing
- Moisture-sensitive materials handling

Coordination & Execution

- Cross-functional R&D coordination
- External partner and vendor management
- Technical documentation and IP development
- Risk identification and tradeoff analysis
- Materials procurement and supplier coordination

PROFESSIONAL EXPERIENCE

FONASE INC. *(Battery Materials Startup)* Chicago, IL
Battery Engineer *Nov 2021 – Present*

- Developed graphene-based Li-ion anode systems from initial formulation through cell-level validation, including electrode design, coating process development, electrolyte evaluation, separator selection, coin cell assembly, and electrochemical performance testing.
- Formulated and iterated novel electrode and electrolyte chemistries that demonstrated significant energy density and operating voltage improvements over conventional Li-ion coin cells in half-cell validation testing.
- Evaluated electrolyte systems including ionic liquid (Pyr14TFSI) with carbonate additives (FEC, ethylene sulfite) and lithium salts (LiTFSI, LiPF6) to optimize SEI formation, interfacial stability, and cycle life.
- Conducted materials characterization (SEM, XRD, FTIR, Raman, XPS, BET) to assess electrode morphology, crystallinity, graphitic structure, doping levels, surface area, and surface chemistry, driving iterative design improvements.
- Analyzed battery aging data – capacity retention, capacity fade, coulombic efficiency, impedance growth, and state of health – across 32+ charge/discharge cycles to inform material and process adjustments.
- Designed electrode architectures with manufacturability in mind, developing a scalable solution-based fabrication process targeting high-energy-density anode integration into next-generation cell architectures.
- Coordinated cell testing and validation with external research partner (Illinois Institute of Technology), managing technical requirements, data exchange, and iterative feedback across institutions.
- Named inventor on granted U.S. Patent No. 12,308,433 covering electrode material composition and fabrication methodology, with 3 additional applications in pipeline (1 notice of allowance, 1 allowable subject matter, 1 continuation).

ELECTRODE FABRICATION AND TESTING AT NORTHWESTERN UNIVERSITY Evanston, IL
R&D Technical Lead (Sponsored Research Collaboration) *Aug 2022 – Dec 2024*

- Performed thin film deposition using Pulsed Laser Deposition (PLD) system at the Materials Research Center shared facility for early graphene deposition experiments, optimizing deposition parameters for target film properties.
- Conducted moisture-sensitive fabrication and materials handling inside GIANTFab inert atmosphere (N2) glovebox facility for thin-film fabrication and testing of air-sensitive electrode materials.
- Executed cleanroom fabrication processes at NUANCE (Nanoscale Characterization Experimental Center), Northwestern's premier nanoscale characterization and fabrication facility.

- Coordinated electrode fabrication and cell testing activities within NUfab cleanroom facilities as part of a sponsored research collaboration.
- Managed procurement of battery materials from external suppliers, scheduling of shared characterization equipment across multiple facilities (NUANCE, NUfab, GIANTFab, Materials Research Center), and coordination with facility staff to execute experimental plans on schedule.
- Analyzed electrochemical aging data including capacity fade profiles, impedance growth trends, and interfacial degradation behavior to inform electrode material selection and process design decisions.

EARLIER EXPERIENCE

Pennsylvania State University — State College, PA
Graduate Research Assistant — *Jan 2021 – May 2021*

- Conducted applied research in heterogeneous catalysis including assay development, metal extraction, and spectroscopic analysis (UV–Vis–NIR, ICP–OES), supporting materials characterization and data-driven evaluation.

National Institute of Technology, Nagpur — Nagpur, India
Research Assistant — *Jan 2018 – May 2018*

- Developed and characterized nanoparticle adsorbents for wastewater treatment, achieving 86.2% adsorption efficiency. Conducted structural and surface analysis (XRD, FTIR, BET).

INDUSTRIAL TRAINEE ROTATIONS

Defense Research and Development Organization (DRDO) — Kanpur, India
Dec 2016 – Jan 2017

- Drafted pilot plant design employing SCADA-based PLCs for producing 100 kg batch polycarbosilane precursor for silicon carbide foam production.
- Designed and sized primary process equipment including batch reactor, shell and tube heat exchanger, and storage vessel, achieving 55% yield.
- Conducted risk analysis on shell and tube bundle failures using TEMA, ASME, and ASTM standards.

Dr. Reddy's Laboratories — Hyderabad, India
Jun 2017 – Jul 2017

- Led basic unit operations project with a 3-member cross-functional team of operators.
- Built simulation environment in Aspen Plus and optimized process parameters, increasing throughput by 15%.

Hindustan Petroleum Corporation Ltd. (HPCL) — Visakhapatnam, India
May 2016 – Jun 2016

- Inspected vacuum and atmospheric distillation column internals for crude distillation operations.
- Surveyed fluid catalytic cracking units (21,000 bbl./day capacity) with commercial process flow diagrams and P&IDs.

Visakhapatnam Steel Plant — Visakhapatnam, India
Dec 2015 – Jan 2016

- Led project on coke oven operations focusing on dry quenching and power generation for blast furnace operations.
- Participated in Kaizen (5S) quarterly audits and reviewed safety hierarchies across continuous casting steel melt shop.

EDUCATION

Harrisburg University of Science and Technology — Harrisburg, PA
Master of Science, Project Management — *Expected June 2026*

Pennsylvania State University — State College, PA
Master of Science, Energy and Mineral Engineering — *May 2021 | GPA: 3.75*

- Non-thesis: Developed an assay to differentiate metal-binding sites on single-atom ceria catalysts

National Institute of Technology — Nagpur, India
Bachelor of Science, Chemical Engineering — *May 2018 | GPA: 3.5*

- Thesis: Adsorption of impurities from effluent water using magnetic nanoparticles

PATENT

U.S. Patent No. 12,308,433 – "Electrode Materials, Electrodes, Devices, and Methods of Making Thereof" (Granted 2025). Named inventor. Covers novel electrode material composition, fabrication methods, and cell architecture for energy storage applications.